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Quick Recap

Apiculture

Introduction

- Apiculture involves the management of bee colonies for the production of honey, beeswax, royal jelly, and other products. Beekeeping has been practiced for thousands of years and is an important agricultural activity in many parts of the world.
- "Apiculture is the scientific method of rearing honeybees."
- Honey bees and their usefulness are known to man from prehistoric times.

Apiary

- It refers to a place where beehives are kept and managed for honey production, pollination services, or other purposes.
- It is essentially a bee farm or bee yard, where the bees live and work.

Beekeeping

- It is the practice of managing bee colonies for the production of honey, beeswax, royal jelly, or other bee products.
- It involves the maintenance of beehives, the care and feeding of bees, and the harvesting of honey and other products.

Apiculture:

- Apiculture is the science and art of beekeeping, which includes the study of bees, their behavior, and their biology, as well as the development and implementation of beekeeping techniques and technologies.

History of bee-keeping

- Development of modern beekeeping has its origin between **1500 and 1851** when many attempts were made to domesticate bees in different types of hives but were not successful because bees attached their combs together as well as to the walls of hive and combs required had to be cut for honey.
- The discovery of the principle of bee space in 1851 by L. L. Langstroth in USA resulted in **first truly movable frame hive**. This bee space was 9.5 mm for Apis mellifera.

Beekeeping in India

- In India first attempt to keep bees in movable frame hives was made in 1882 in Bengal and then in 1883-84 in Punjab.
- In south India, Rev. Newton during 1911-1917 trained several beekeepers and devised a hive for indigenous bee *Apis cerana* based on principle of bee space (which was named after his name as "**Newton hive**").
- Beekeeping was also started in the Travancore state (now Cochin) in 1917 and in Mysore in 1925.
- In Himachal Pradesh modern beekeeping with indigenous honey bee *A. cerana* started in 1934 at Kullu and in 1936 at Kangra.
- The exotic bee *A. mellifera* was successfully introduced for the first time in India in **1962 at Nagrota Bagwan** (then in Punjab state and now in Himachal Pradesh), because this bee has potentials to produce more honey.

Species of honey bee:

1. Indian bee: *Apis cerana indica.*
2. Rock bee: *Apis dorsata.*
3. European bee: *Apis mellifera*
4. Little bee: *Apis florea*
5. Indian Black Honey bee: *Apis karinjodian*
6. Dammer bee or stingless bee: *Melipona iridipennis*

1. ***Apis dorsata*:-**
    - They construct single comb in open (About 6ft long and 3ft deep)
    - They shift the place of the colony often.
    - Rock bees are ferocious and difficult to rear.
    - They produce about 37 Kg honey /comb/year.

2. ***Apis florea*:**
    - They also construct comb in open of the size of palm in branches of bushes, hedges, buildings, caves, empty cases etc.
    - They produce about 1/2Kg honey/year/hive.
    - They are not rearable as they frequently change their palce.
    - The size of the bees is smallest among 4 *Apis Sp*. described.

3. ***Apis cerana indica* (Indian bee/Asian bee):**
    - They make multiple parallel combs on trees and cavities in darkness.
    - The bees are larger than *Apis florae* but smaller than *Apis mellifera.*
    - They produce about 2- 5 Kg of honey/year/hive.
    - They are more prone to swarming and absconding.
    - They are native of India/Asia.

4. ***Apis mellifera* (Italian bee or European bee):**
    - They also make multiple parallel combs in cavities in darkness.
    - They are larger than Indian bees but smaller than Rock bees.
    - They have been imported from European countries.(Italy)
    - They yield on an average 35 Kg/hive/year.
    - First introduced to PAU(Punjab Agriculture University) in 1963 by **A.S. Atwal.**
    - Superior than Indian bee in foraging & honey gathering abilities with prolific queen.

5. ***Apis karinjodian* (Indian Black Honeybee):**
    - A new species of endemic honeybee has been discovered after a gap of more than 200 years.
    - The Indian black honeybee, ranges from the central Western Ghats and Nilgiris to the southern Western Ghats, covering the States of Goa, Karnataka, Kerala and parts of Tamil Nadu.
    - The species has been classified as near threatened (NT) in the State based on the IUCN Red List Categories and Criteria.

6. ***Melipona iridipennis* (Dammer Bee):**
    - These bees are much smaller than the true honey bees and build irregular combs of wax and resinous substances in crevices and hollow tree trunks.
    - **The stingless bees** have the importance in the pollination of various food crops.
    - They bite their enemies or intruders.

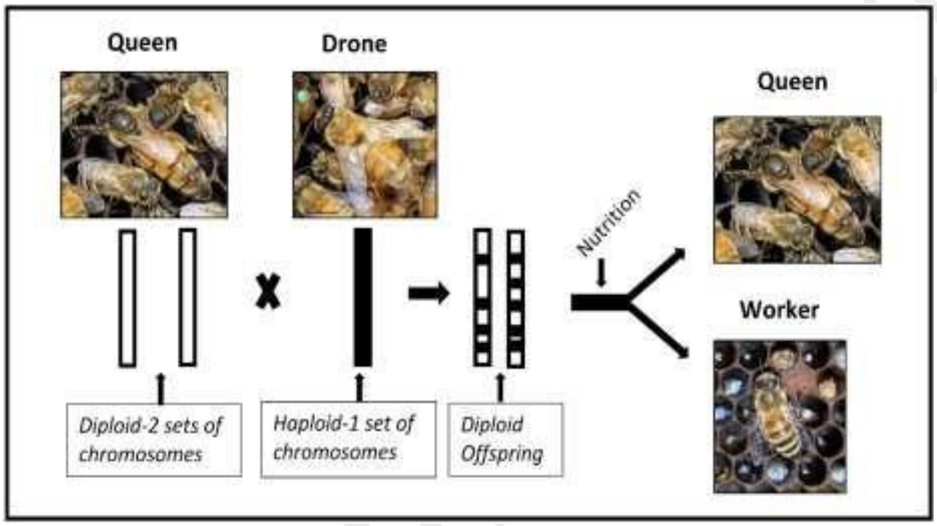
Honey Bee Castes:

- Every honey bee colony comprises of a single queen, a few hundred drones and several thousand worker castes of honey bees.
- Queen is a fertile, functional female
- Worker is a sterile female and
- The drone is a male insect.

Queen:

- There is only one queen in a colony.
- It is considerably larger than the members of other castes.
- Because of her long tapering abdomen, it appears more wasp-like than other inmates of the colony.
- The queen is the **only individual which lay eggs** in a colony and is the mother of all bees.
- It lays upto 2000 eggs per day in *Apis mellifera*.
- When her spermatheca is filled with sperms, she will start laying eggs. She lives for approximately 3 years.
- The secretion from mandibular gland of the queen is called **queen's substance**. The queen substance if present in sufficient quantity, prevent swarming and absconding of colonies, prevent development of ovary in workers, and maintains colony cohesion.
- The differentiation in worker and queen is due to the quantity and quality of food fed to the larva.
- The larva which becomes the queen is fed the royal jelly, a secretion from **hypopharyngeal glands** of the worker bees.
- The queen is reared in large finger-shaped cells in the lower portion of the combs.
- Only one queen can remain in a colony, but during unfavourable season two queens are also observed.

- The old queen is killed as soon as the new queen is fertilized.
- Generally queens are reared only during swarming season, but if the queen dies accidentally the bees can rear a new queen.
- The phenomenon of raising queen in off-season is called **supersedure**
- If the food is scarce, workers do not permit the queen to lay eggs.
- The queen is carefully, looked after by young workers, known as-attendants, which feed her and keep her clean and well combed. The queen never leaves the, hive except with a swarm.



Drone:

- The drones are the **male bees**.
- They have no sting; a suitable proboscis for gathering nectar is also absent. They are, therefore, physically incapable for the ordinary work of the hive.
- Their only function is **to impregnate the young queen** a task which they are unable to perform until they are about 10 days of age. They also help in maintenance of hive temperature.
- They go out of the hive only at the mid-day when the weather is warm. The number of drones in a colony often is very large amounting to hundreds and sometimes to thousands.
- The drones are reared and tolerated during the breeding season.
- They are driven out of the hive to die of starvation before the monsoon and the winter.
- The drones are produced by **unfertilized eggs** of the queen, or by those workers which take up the reproductive function due to the absence of a queen in a colony.
- The normal life-span of a drone is 57 days.
- Mating takes place in the open when the queen is in flight. The drone dies in the act or immediately afterwards. Its abdomen has to burst open to allow the genital organ to function

Worker Bee

- The workers are the smallest inhabitants of the beehive.
- They form the bulk of the population. The number of workers in a colony varies from 1,500 to 50,000.
- They are **imperfect females** incapable of laying eggs.
- On certain occasions when the colony is in need of a queen, some of the workers start laying eggs from which only drones are produced. These workers, called **laying workers**, are killed as soon as a new queen is introduced or produced in the colony.
- The life-span of a worker is about 4 weeks during active season and 8 to 10 weeks during less active season.
- Their range of flight varies from 1,000 to 1,500 m.
- The division of work within a colony among the worker bees is based on the age of the individual and on the needs of the colony.
- Normally, the young bees, immediately after their emergence, do the work of cleaning cells and feeding older larvae.
- When they are grown and their hypopharyngeal glands have developed, they secrete the royal jelly with which they feed the younger larvae. These bees are called **nurse bees.**
- For the first 2 to 18 days of their life, the bees perform indoor duty inside the hive, including comb construction when some young bees start secreting wax. Later on they become foragers, collect water, pollen, nectar and propolis (bee-blue).
- Thus the lifespan of workers can be divided into two phases as first three weeks for house hold duty and rest of the life for outdoor duty.

House hold duty includes:

- Build comb with wax secretion from wax glands.
- Feed the young larvae with royal jelly secreted from hypopharyngeal gland.
- Feed older larvae with bee-bread (pollen+ honey)
- Feeding and attending queen.
- Feeding drones.
- Cleaning, ventilating and cooling the hive.
- Guarding the hive.
- Evaporating nectar and storing honey.

Outdoor duties

- Collecting nectar, pollen, propolis and water.
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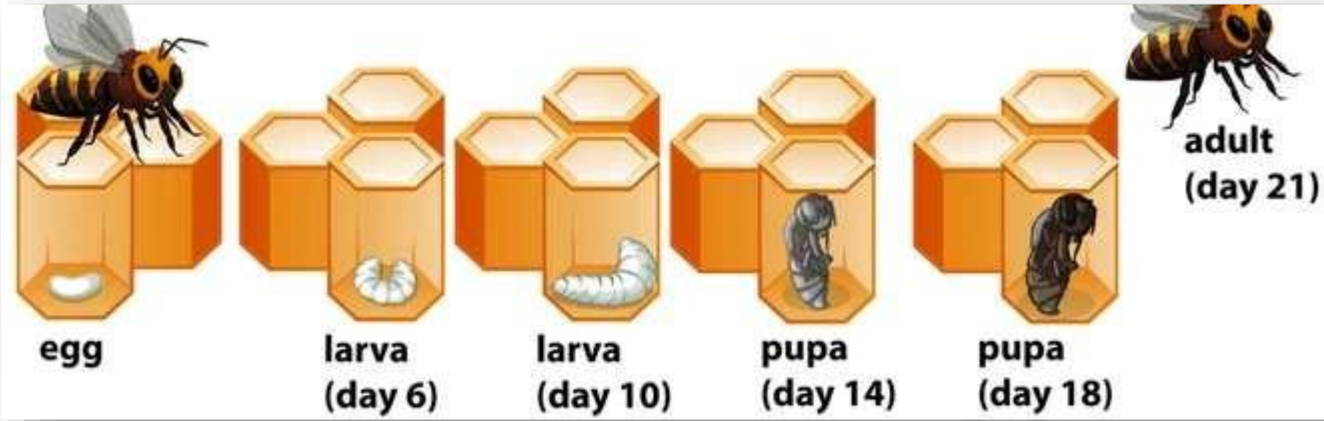
Life Cycle of Bee:



The life cycle of honey bees is divided into four stages: the egg, the larval, the pupal and the adult stage.

Stage 1 – The Egg Stage:

- Queen bee is the only bee in the colony who is capable of laying about 2,000 to 3,000 eggs in one day.
- The queen bee lays both fertilized egg and unfertilized egg. The fertilized egg develops into female bees or queen bees. The unfertilized egg hatches and male bees are born; also known as drone bees.

Stage 2 – The Larval Stage:

- The difference between a worker and the queen bee is made three days after the egg transforms into larvae and six days after the egg is laid in the beehive.
- The "royal jelly" is fed to all the larvae, i.e., the female bees, the workers and the drone bees during their initial three days as larvae.
- The larva sheds skin multiple times throughout this stage.
- Later, the royal jelly is fed only to the female larvae, which eventually becomes a queen bee. Finally, the worker bees cover the top of the cell with beeswax to protect and facilitate the transformation of the larvae into a pupa.

Stage 3 – The Pupal Stage:

- Here the bee has developed parts like wings, eyes, legs and small body hair that physically appears close to an adult bee.

Stage 4 – The Adult Stage:

- Once the pupa is matured, the new adult bee chews its way out of the closed-cell.
- The queen bee takes 14-16 days from the egg stage to form into an adult.
- The worker bee takes 18 to 22 days for complete development, and drone bees take 24 days to develop into an adult bee.

Bee products:

Honey:

- Honey is a sweet, viscous substance produced by honeybees from the nectar of flowers.
- It is a natural sweetener and can be used as a substitute for sugar in many recipes.
- It also has antibacterial properties and can be used to treat minor wounds and sore throats.

Beeswax:

- Beeswax is a natural wax produced by honeybees.
- It has many uses, including as a natural skincare ingredient, as a lubricant for tools and machinery, and as a component in candles and other products.

Propolis:

- Propolis is a sticky resin-like substance produced by honeybees from tree sap.
- It has antibacterial, antifungal, and antiviral properties and is used in some natural remedies and healthcare products.

Royal jelly:

- Royal jelly is a milky substance produced by honeybees to feed their larvae and queen bee.
- It is high in nutrients and is sometimes used as a dietary supplement or skincare ingredient.

Pollen:

- Pollen is the male reproductive cells of plants that are collected by bees as a food source.
- It is high in protein, vitamins, and minerals and is sometimes used as a dietary supplement.

Important Terminologies:

1. Swarming:

- Swarming is a natural method of colony multiplication in which a part of the colony migrates to a new site to make a new colony.
- It occurs when a colony builts up a considerable strength or when the queen's substance secreted by queen falls below a certain level.
- Swarming is a potent instinct in bees for dispersal and perpetuation of the species.

2. Supersedure:

- When a old queen is unable to lay sufficient eggs, she will be replaced or superseded by supersedure queen.
- Or when she runs out of spermathezoa in her supermatheca, and lays many unfertilized eggs from which only drones emerge.
- In this case, one or 2 queen cells are constructed in the middle of the comb and not at the bottom.
- At a given time both new and old queens are seen simultaneously. Later the old queen disappears.

3. Emergency queen:

- In the event of death of the queen the eggs (less than 2½ days old) in worker cells are selected and the cell extended like a queen cell.
- It is fed with abundant royal jelly and covered into queen.
- The first queen which comes out of the emergency queen cells kills other stages of queen inside the cells and then go for mating.

4. Communication in bees:

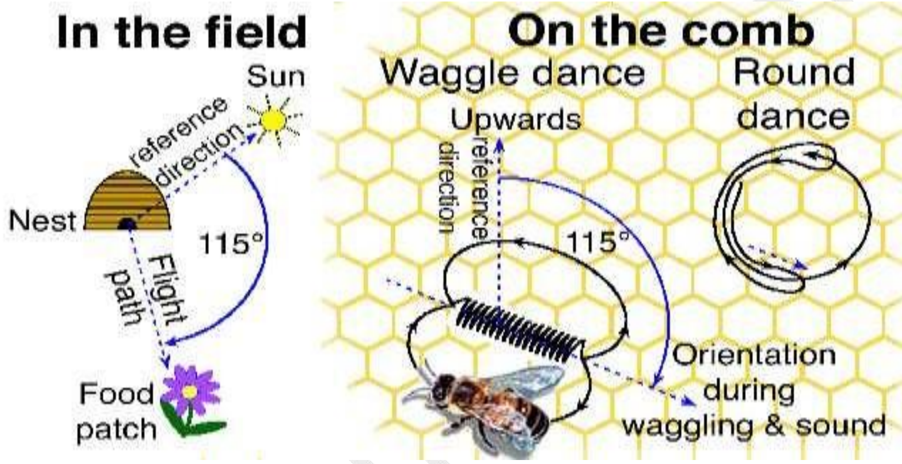
- Bees communicate using various phenomones, including the queen's substance, alarm pheromone emitted from sting and secretion of tarsal gland.
- In addition the bees also communicate by performing certain dances.
- When scout bees return to the box after foraging they communicate to the other forages present in the box about the direction and distance of the food source from the hive by performing dances.

1. Round Dance:

- Round dance is used to indicate a short distance **(Less than 50m in case of *A.mellifera***).
- The bee runs in circles, first in one direction and then in opposite direction, (clockwise and anticlockwise).

2. Tail wagging dance or Wag-tail dance.

- This is used to indicate long distance. (**more than 50m**).
- Here the bee makes two half circles in opposite directions with a straight run in between.
- During the straight run, the bee shakes (wags) its abdomen from side to side, the
- number of wags per unit time inversely proportional to the distance of the food (**more the wags, less the distance).**
- The direction of food source is conveyed by the angle that the dancing bee makes between its straight run and top of the hive which is the same as between the direction of the food and direction of the sun.



5. Hive:

- A structure where honeybees live and store honey.
- It is typically made up of a bottom board, brood boxes, honey supers, frames, and a lid.

6. Colony:

- A group of honeybees living together in a hive.
- It typically consists of a queen bee, worker bees, and drones.

7. Comb:

- The hexagonal wax structure that bees build inside the hive to store honey, pollen, and raise brood.

8. Brood:

- The young bees that are raised in the hive, including eggs, larvae, and pupae.

9. Nectar:

- A sweet liquid that bees collect from flowers and use to make honey.

10. Pollen:

- A powdery substance that bees collect from flowers as a protein source.

11. Honeycomb:

- The waxy structure created by bees to store honey.

12. Honey extractor:

- A machine used to extract honey from the comb.

13. Smoker:

- A tool used to calm bees by blowing smoke into the hive.

14. Absconding:

- In beekeeping, it refers to the sudden departure of an entire colony of honeybees from their hive.
- This can happen when the bees decide to abandon the hive due to various reasons such as disease, pests, lack of food, or environmental factors.
- Absconding is different from swarming, which is a natural process where a colony splits into two, and the queen and a portion of the bees leave the original hive to create a new one.

15. Nuptial Or Marriage Flight:-

- A term used in beekeeping and entomology to describe the flight of a queen bee and the mating process with drones.
- Nuptial flight is an essential part of the honeybee life cycle, as it allows the queen bee to mate with multiple drones and ensure genetic diversity in the colony.

16. Bee Box structure:-

1. **Bottom board:** This is the base of the box and serves as the foundation for the rest of the hive. It provides a flat surface for the box to rest on and serves as the entrance for the bees.

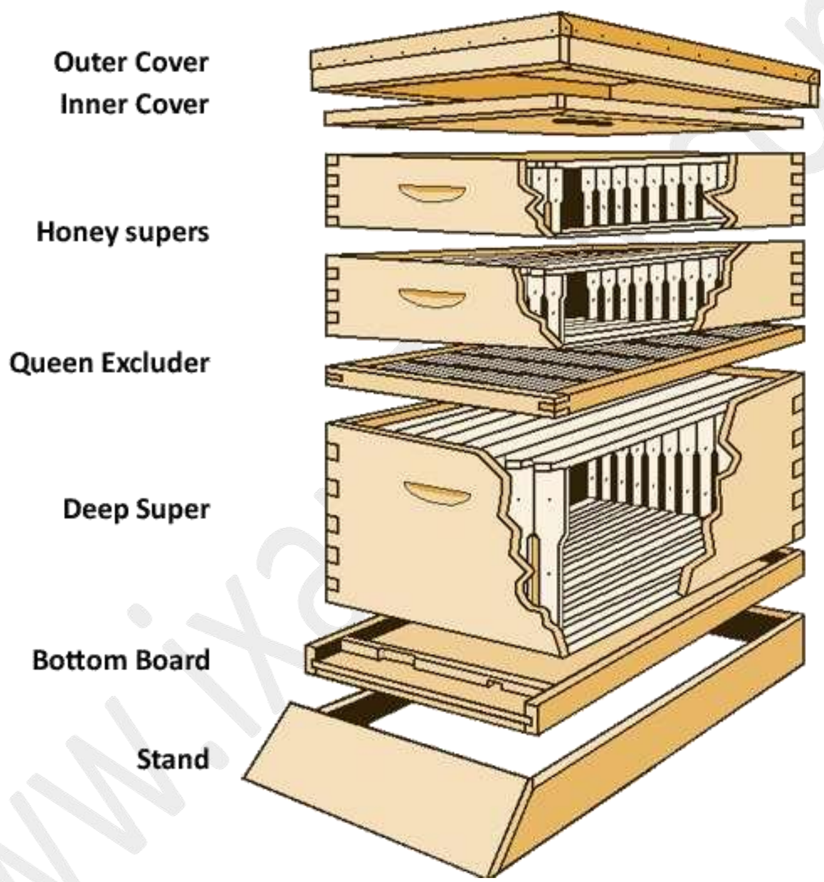
2. **Brood box:** This is the main section of the hive where the bees raise their young. It typically consists of one or more boxes stacked on top of each other, each containing frames where the bees build comb and raise brood.

3. **Frames:** These are wooden frames that hold the wax foundation for the bees to build comb on. They are typically made of wood and have a plastic or wax foundation that the bees will build their comb on.

4. **Super:** This is a smaller box that sits on top of the brood box and is used for honey storage. It is typically smaller than the brood box and contains frames for honey storage.

5. **Inner cover:** This is a flat board that sits on top of the uppermost super and provides insulation for the hive. It also has a small hole in the center that serves as a ventilation hole.

6. **Outer cover:** This is the outermost layer of the hive and provides protection from the elements. It is typically a wooden cover with a metal or plastic covering.



Pest and Diseases of Bees

Pests of Honey Bees:

1. **Wax Moths (*Galleria mellonella, Achroia grisella, Achroi innotata*)**:
   Symptoms: Feeding on propolis, pollen, and wax in combs. They create silken tunnels, leaving behind debris and webbing, which weakens colonies.
   Control: Regular hive examination, removal of debris, fumigation, and storing spare combs in sealed containers.

Adult Greater wax moth and Galleries Formed on Bee Comb

2. **Ants (*Camponotus compressus, Dorylus labiatus, Monomorium spp.):***
   Threat: Attacking weak colonies, stealing honey, pollen, and brood.
   Control: Using ant pans, oil bands, or treating ant nests with pesticides.

Camponotus compressus

3. **Predatory Wasps**
   **(*Vespa cincta, V. velutina, V. magnifica, V. tropica,*
   V. basalis, *Palarus orientalis, Phyllanthus ramakrishnae V*):**
   Behavior: Capturing bees at hive enterance for feeding their young.

Prevention: Narrowing hive entrances, destroying wasp nests by burning.

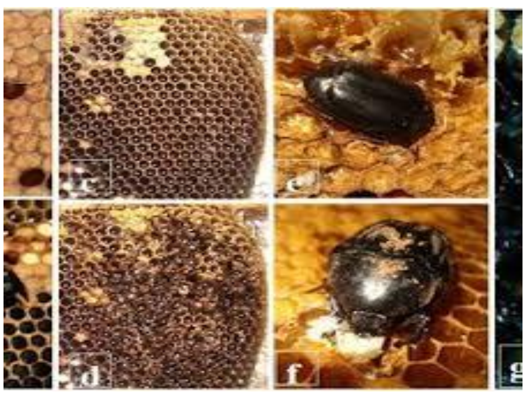
Scientists insert a hornet into a nest of Japanese honeybees - and the tiny creatures swarm over the huge predator to 'cook' it inside a 'bee ball'

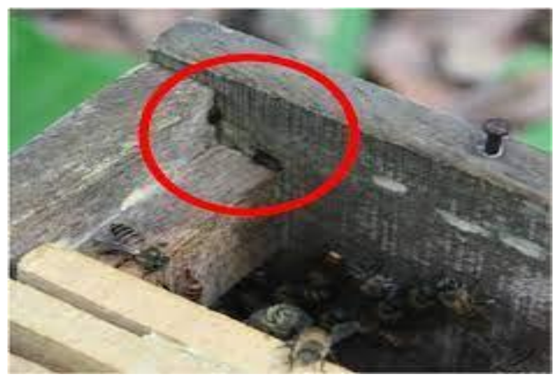
The creatures swarm over the hornet, encasing it in a mass of bee bodies which slowly 'cooks' the inch-long predator

4. **Wax Beetles (*Platybolium alvearium*):**

Habit: Feeding on debris and old combs in weak colonies.

Control: Regular comb examination, cleaning bottom boards. Kill the fecunded females by flapping





5. Birds:

- King Crow (*Dicrucus spp.*)
- Bee Cavers (*Merops spp.*)

Dicrucus spp.

Merops spp.

6. Mites

- **Tracheal Mites: *Acarapis woodi*:** They feed on heamolymph by piercing through the tracheal wall.
- **Ectoparasitic Mites**: These mites causes severe damage to *A. mellifera* colonies.
    - ***Tropilaelaps clareae*** : Symptoms of attack appears as irregular brood pattern, perforated brood capping, malformed or dead bees at hive entrance.
    - For control Sulphur dusting @200mg/frame
    - ***Varroa destructors****:* Mites reproduces and grows in sealed brood cells by feeding on hemolymph of bee pupa.
    - For control Formic acid fumigation @50ml/hive in sponge pads covered with perforated polythene bags

Acarapis woodi

Tropilaelaps clareae

Varroa destructors

Bee louse, *Braula coeca*: These wingless flies sit near thorax of bee and feeds on food material by coming near to opening of salivary glands at mouth.

Braula coeca

Diseases of Honey Bees:

1. **Nosema Disease (*Nosema apis*):**

- Caused by a protozoan, resulting in dysentery and weakened flight during cold weather.
- Control involves using Fumagillin, an antibiotic, administered via sugar syrup.

- **European Foul-Brood:** Caused by *Streptococcus pluton*, **affecting larvae and leading to foul odor.** Controlled by Terramycin and quarantine measures.

- **American Foul Brood:** Caused by Bacillus, affecting larvae, leading to putrefaction **and fishy odor.** Controlling involves destroying infected colonies.

- **Sac-Brood Disease and Thai Sac-Brood Virus:** Virus diseases affecting larvae, **causing sac-like appearance in some cases**. Prevention involves isolation and requeening.

- **Chalk Brood and Stone Brood:** Fungal diseases affecting larvae, causing **mummification**. Chalk brood is caused by *Ascosphaera apis*.